Fusão de Informação em Análise de Dados
Information Fusion in Data Analysis

Project - I

Objective: *The aim is to apply multidimensional Kalman filters in the fusion of sensor data in nonlinear systems, Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF), to obtain the estimate of a robot's location.*

Introduction

This project aims to apply Kalman filters (EKF and UKF) to the localization of a robot, which is represented by a non-linear model. The robot is considered to have a speed sensor, a gyroscope and a GNSS (Global Navigation Satellite System) sensor.

Background

By applying the Kalman filters (EKF and UKF) to the fusion of sensor data from a given robot, the aim is to obtain, by simulating its location, a result similar to that represented in Figure 1.

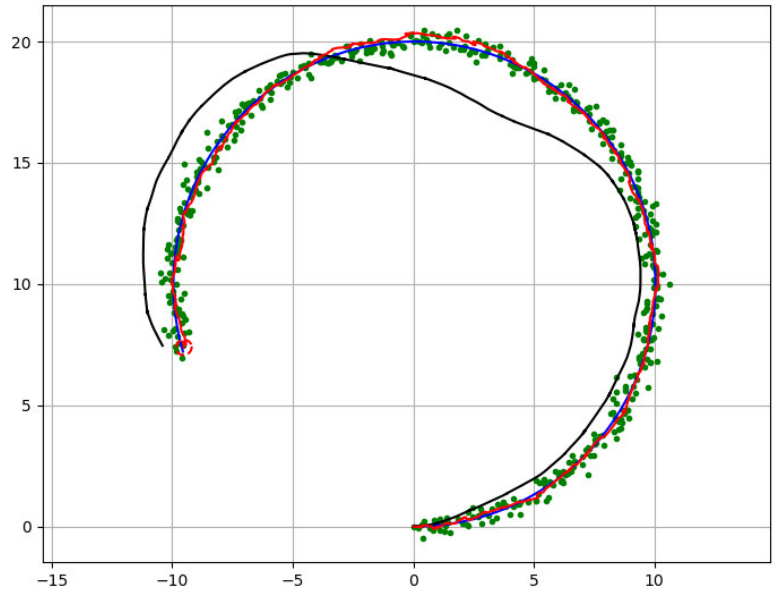


Figure 1. Representation of a robot's trajectory, used to estimate its location with Kalman Filters (EKF or UKF).

In this figure, the blue line is the true trajectory, the black line is the trajectory by relative positioning (dead-reckoning), the green points correspond to the observation of the robot's positioning (e.g. with GPS) and the red line is the trajectory estimated with the EKF or UKF. The dotted red ellipse shown at each iteration represents the estimated covariance ellipse.

- *Dynamic Model of the robot*

In this simulation, the robot state vector has the following 4 states at time *k*:

$$\boldsymbol{x}_k = \begin{bmatrix} x_k \\ y_k \\ \phi_k \\ v_k \end{bmatrix}$$

where *x* and *y* are the 2D *x*-*y* position of the robot, ϕ is the orientation and *v* is the velocity.

Given that the robot has a speed sensor and a gyroscopic sensor, the input vector at each instant *k* is represented by:

$$\boldsymbol{u}_k = \begin{bmatrix} v_k \\ \omega_k \end{bmatrix}$$

In addition, the robot has a GNSS sensor, allowing it to observe its *x*-*y* position at each instant of time *k*:

$$\boldsymbol{z}_k = \begin{bmatrix} x_k \\ y_k \end{bmatrix}$$

The input and observation vectors are considered to include noise associated with the respective sensors.

The robot model can be represented by:

$$\dot{\phi} = \omega$$

Thus, the dynamic nonlinear model of the robot in the dynamic state space is given by:

$$\boldsymbol{x}_{k+1} = \boldsymbol{A}\boldsymbol{x}_k + \boldsymbol{B}\boldsymbol{u}_k + \boldsymbol{w}_k$$

where:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \boldsymbol{B} = \begin{bmatrix} \cos(\phi)\Delta t & 0 \\ \sin(\phi)\Delta t & 0 \\ 0 & \Delta t \\ 1 & 0 \end{bmatrix}$$

and Δt is the time step.

The corresponding Jacobian matrix results in:

$$\boldsymbol{J_A} = \begin{bmatrix} \frac{\partial x}{\partial x} & \frac{\partial x}{\partial y} & \frac{\partial x}{\partial \phi} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial x} & \frac{\partial y}{\partial y} & \frac{\partial y}{\partial \phi} & \frac{\partial y}{\partial v} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial \phi} & \frac{\partial \phi}{\partial v} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial \phi} & \frac{\partial v}{\partial v} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -v\sin(\phi)\Delta t & \cos(\phi)\Delta t \\ 0 & 1 & v\cos(\phi)\Delta t & \sin(\phi)\Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The robot can obtain information about its *x*-*y* position from the GNSS sensor, obtaining the following observation model:

$$\boldsymbol{z}_k = \boldsymbol{C}\boldsymbol{x}_k + \boldsymbol{r}_k$$

where:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

The respective Jacobian matrix is given by:

$$\boldsymbol{J_C} = \begin{bmatrix} \frac{\partial x}{\partial x} & \frac{\partial x}{\partial y} & \frac{\partial x}{\partial \phi} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial x} & \frac{\partial y}{\partial y} & \frac{\partial y}{\partial \phi} & \frac{\partial y}{\partial v} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Project Tasks:

Considering the presented framework, perform the following tasks (in MATLAB or Python)::

1. Create a function that implements the robot's dynamic model;
2. Create a function that implements the observation model;
3. Create a function that represents the acquisition of data from the sensors;
4. Create a function that implements the Extended Kalman Filter (EKF);
5. Create a function that implements the Unscented Kalman Filter (UKF);
6. Specify the various parameters of the robot's filters and simulation;
7. Simulate the robot's movement and dynamically represent its location, considering the trajectories and ellipse indicated in Figure 1;
8. Compare the performance of the different approaches using appropriate indicators (Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Coefficient of Determination (R-squared), …).
9. Comment on the results obtained.

Organization and Materials to be delivered:

The Project must be developed in groups of two elements.

The group can choose another dynamic system, which must be previously approved by the teacher.

In addition to the code, you must deliver a report (you can use a Jupyter Notebook to integrate the code and the report) where you indicate the main steps of applying each of the filters and present and discuss comparatively the results obtained.

The project report and code must be submitted to inforestudante until the stipulated deadline.
The project grade will result from the assessment of the code and the report, as well as the corresponding discussion.